Closing the center-embedding gap in Spanish: can grammatical features do the work of memory?

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Abstract

Sentences with multiple center-embedded clauses are famously hard to understand (Karlsson 2007). In doubly embedded relative clauses, this difficulty often produces the missing-VP illusion: when the second of three required verbs is missing (see examples below), readers rate the ungrammatical sentence as more acceptable than its grammatical counterpart (Lewis and Vasishth 2005; Hahn et al. 2022). Pañeda and Lago (2024) recently documented this illusion in Spanish. They found that the size of the effect depended on how rarely the verb sequence occurred in the language, while individual working-memory span had no effect on acceptability ratings. This pattern points to interference during cue-based memory retrieval as a primary source of the illusion.

Spanish allows a direct test of this retrieval account because it has three relativizers that appear in the same syntactic position yet differ in morphology. The default relativizer *que* is invariant and relies on an accompanying article to show gender and number (*a la que / al que*). By contrast, *el cual* inflects for gender and number on the pronoun itself (*a la cual / al cual*), and *quien* marks number while restricting its antecedent to human referents. Under cue-based retrieval models (Lewis and Vasishth 2005), the missing-VP illusion arises when noisy memory representations of the clause heads are confused at retrieval. If that is right, overtly inflected relativizers should make the encoding of each clause more distinct and so weaken the illusion, with clause length and embedding depth held constant.

We test this hypothesis with 120 native Spanish speakers across the six conditions shown below. Participants give speeded acceptability judgments while we record their eye movements during reading, along with scalp EEG. The more grammatical features the relativizer encodes, the fewer missing-VP illusions we expect in the behavioral ratings and shorter regression-path times at the final verb. In the ERP record, more distinctive relativizers should reduce the sustained anterior negativity seen while open dependencies are held in memory and restore the P600 response to the missing verb. Together, these measures test whether the morphological content of function words helps the parser keep nested syntactic dependencies apart.

Keywords: sentence processing; cue-based retrieval; working memory; morphosyntactic features; ERP; Spanish

SUPPORTING MATERIALS

All conditions are double center embeddings of identical depth; the featural content of the relativizer is crossed with grammaticality (VP2 entrevistó present vs. deleted).

(1) a. La política **a la que** el periodista **al que** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la que** el periodista **al que** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
In this baseline condition with invariant *que* (used in all prior Spanish work), gender and number are marked only on the article.

(2) a. La política **a la cual** el periodista **al cual** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a la cual** el periodista **al cual** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
With *el cual*, gender and number are inflected on the relative pronoun itself in addition to the article.

(3) a. La política **a quien** el periodista **a quien** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]

b. *La política **a quien** el periodista **a quien** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]

With *quien*, the pronoun inflects for number and restricts the antecedent to a human referent.

Differential object marking (*a*) prevents a subject-relative reading in all conditions, so no sentence is locally ambiguous at the relativizer.

REFERENCES

Hahn, Michael et al. (2022). "A resource-rational model of human processing of recursive linguistic structure". In: *Proceedings of the National Academy of Sciences* 119.43, e2122602119. DOI: 10.1073/pnas.2122602119.

Karlsson, Fred (2007). "Constraints on multiple center-embedding of clauses". In: *Journal of Linguistics* 43.2, pp. 365–392. DOI: 10.1017/S0022226707004616.

Lewis, Richard L. and Shravan Vasishth (2005). "An activation-based model of sentence processing as skilled memory retrieval". In: *Cognitive Science* 29.3, pp. 375–419. DOI: 10.1207/s15516709cog0000_25.

Pañeda, Claudia and Sol Lago (2024). "The missing VP illusion in Spanish: Assessing the role of language statistics and working memory". In: *Open Mind: Discoveries in Cognitive Science* 8, pp. 42–66. DOI: 10.1162/opmi_a_00118.